FACULTY OF INFORMATION AND COMMUNICATION TECHNOLOGY

ITDS 294
Fundamentals of Electronic Commerce Systems

Session 3 : E-Commerce Infrastructure

Session 3

E-Commerce Infrastructure →

โครงสร้างของ Internet และการทำงานของ WWW รวมทั้งคุณลักษณะและ Service ที่สนับสนุน E-Commerce ทำความเข้าใจผลกระทบของ Mobile Application

In Class Activity 1 : Building Our Own E-Commerce System VS Using E-Commerce Solution →

ให้นักศึกษาอภิปรายร่วมกันในเรื่องของการสร้าง E-Commerce ขึ้นมาใช้เอง หรือการใช้ E-Commerce แบบสำเร็จรูป หรือการขายสินค้าผ่าน Marketplace ผ่านบอร์ดเกม E-Commerce Infrastructure

Learning Outcome for Session 3

E-Commerce Infrastructure

นิสิตสามารถอธิบายคุณลักษณะและบริการของ Internet และ WWW ที่สนับสนุนการทำงาน E-Commerce

In Class Activity 1 : Building Our Own E-Commerce System VS Using E-Commerce Solution

นิสิตสามารถแสดงความคิดเห็นเกี่ยวกับทางเลือกในการสร้าง E-Commerce ที่มีความเหมาะสมกับลักษณะของธุรกิจ และสถานการณ์ต่าง ๆ ได้

E-Commerce Infrastructure

โครงสร้างของ Internet และการทำงานของ WWW รวมทั้งคุณลักษณะที่สนับสนุน E-Commerce ทำความเข้าใจผลกระทบของ Mobile Application

The Internet :
Technology Background

The **Internet** is an interconnected network of thousands of networks and millions of computers (sometimes called *host computers* or just *hosts*), linking businesses, educational institutions, government agencies, and individuals.

The Internet provides 4 billion people around the world with services such as e-mail, apps, newsgroups, shopping, research, instant messaging, music, videos, and news.

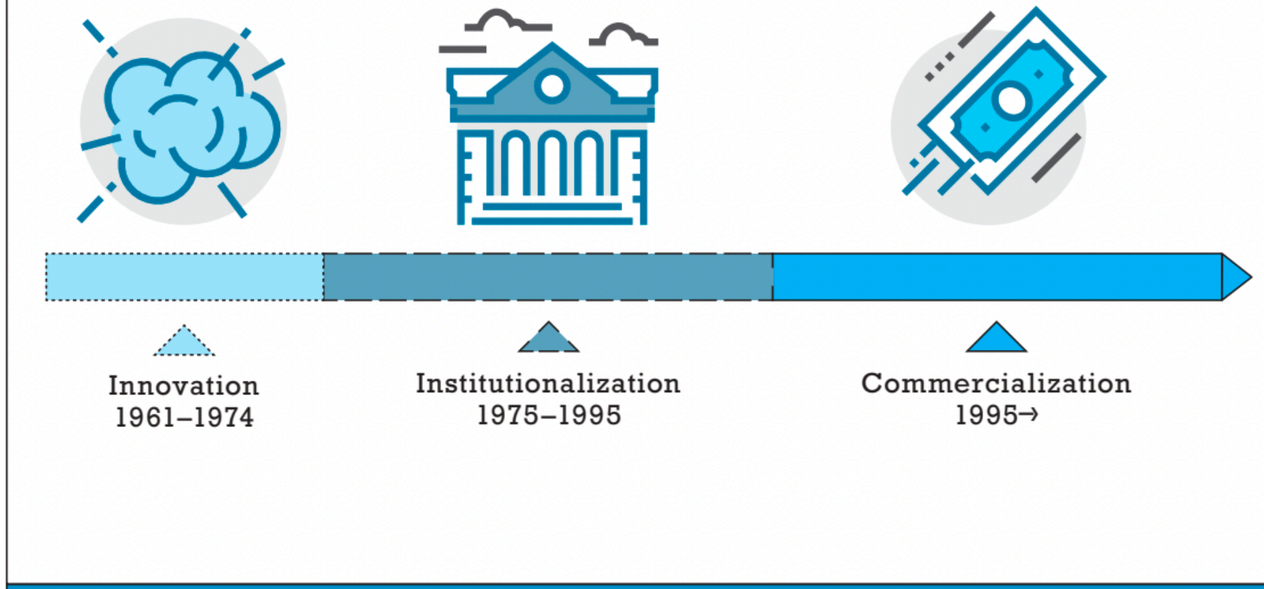
The Internet :
Technology Background

The **Web** is one of the Internet's most popular services, providing access to billions, perhaps trillions, of web pages, which are documents created in a programming language called HTML that can contain text, graphics, audio, video, and other objects, as well as "hyperlinks" that permit users to jump easily from one page to another. Web pages are navigated using web browser software.

The Evolution of The Internet

The history of the Internet can be segmented into three phases

FIGURE 3.1 **STAGES IN THE DEVELOPMENT OF THE INTERNET**



The Internet has developed in three stages from 1961 to the present. In the Innovation stage, basic ideas and technologies were developed; in the Institutionalization stage, these ideas were brought to life; in the Commercialization stage, once the ideas and technologies had been proven, private companies brought the Internet to millions of people worldwide.

The Evolution of The Internet

YEAR	EVENT	SIGNIFICANCE
INNOVATION PHASE 1961–1974		
1961	Leonard Kleinrock (MIT) publishes a paper on "packet switching" networks.	The concept of **packet switching** is born.
1962	J. C. R. Licklider (MIT) writes a memo calling for an "Intergalactic Computer Network."	The vision of a global computer network is born.
1969	The first packet-switched message is sent on ARPANET from UCLA to Stanford.	The communications hardware underlying the Internet is implemented for the first time.
1972	E-mail is invented.	The first "killer app" of the Internet is born.
1973	Bob Metcalfe (Xerox PARC Labs) invents Ethernet and local area networks (LANs).	Ethernet and LANs enable the development of **client/server computing**.
1974	"Open architecture" networking and TCP/IP concepts are presented in a paper by Vint Cerf (Stanford) and Bob Kahn (BBN).	The paper provides the conceptual foundation for a single common communications protocol, **TCP/IP**, that could connect disparate LANs and computers, and a common addressing scheme for all computers connected to the network.

The Evolution of The Internet

INSTITUTIONALIZATION PHASE 1975–1995		
1977	Lawrence Landweber envisions CSNET (Computer Science Network), a network for U.S. university and computer industry research groups.	CSNET is a major milestone on the path to the development of the global Internet.
1980	TCP/IP is officially adopted as the U.S. Department of Defense (DoD) standard communications protocol.	The DoD, the single-largest computing organization in the world at the time, validates TCP/IP.
1981	IBM introduces IBM PC, its first personal computer.	Personal desktop computers provide the foundation for access by millions of people to the Internet.
1984	Apple releases the HyperCard program as part of its Macintosh operating system.	The concept of "hyperlinked" documents that permit a user to jump from one page to another is commercially introduced.
1984	Domain Name System (DNS) is introduced.	DNS provides a user-friendly system for translating IP addresses into words that people can easily understand.
1989	Tim Berners-Lee (CERN) proposes a worldwide network of hyperlinked documents based on a common markup language called HTML—HyperText Markup Language.	The concept of an Internet-supported service called the **World Wide Web** based on **HTML** is born.
1993	Mosaic, the first graphical web browser, is invented.	Mosaic makes it easy for ordinary users to connect to HTML documents anywhere on the Web.
1994	Netscape, the first commercial web browser, becomes available; the first banner advertisements appear on Hotwired.com in October 1994.	The beginning of **e-commerce**.

The Evolution of The Internet

COMMERCIALIZATION PHASE 1995–PRESENT		
1995	Major telecommunications companies take over operation of the Internet backbone. Network Solutions (a private firm) is given a monopoly to assign Internet addresses.	The **fully commercial civilian Internet** is born.
1995	Jeff Bezos founds Amazon.	E-commerce begins in earnest.
1998	The Internet Corporation for Assigned Names and Numbers (ICANN) is created.	Governance over domain names and addresses passes to a private, nonprofit, international organization.
2007	The Apple iPhone is introduced.	The **iPhone** represents the beginning of the development of a viable **mobile platform**.
2008	Internet "cloud computing" becomes a billion-dollar industry.	Internet capacity is sufficient to support on-demand computing resources as well as software applications (**cloud computing**).
2011	ICANN expands the domain name system.	ICANN expands generic top-level domain names from about 300 to potentially thousands.
2012	World IPv6 Launch day.	IPv6 increases the pool of available Internet addresses, enabling continued expansion of the Internet.
2013	The Internet of Things (IoT) starts to become a reality.	Internet technology spreads beyond computers and mobile devices to anything that can be equipped with sensors, paving the way for an **Internet of Things (IoT)**.
2014	Apple introduces Apple Pay and Apple Watch.	Apple Pay becomes the first widely adopted mobile payment system; Apple Watch ushers in a new era of wearable Internet-connected technology.
2019	The commercial availability of 10 Gbps Internet access increases.	Advanced technologies such as virtual reality, augmented reality, artificial intelligence, and 4k streaming video drive demand for faster broadband access speeds.
2020	The Covid-19 pandemic sweeps the globe; Internet usage skyrockets.	The pandemic illustrates the resiliency of the Internet to cope with a tremendous surge in demand.
2021	Facebook rebrands as Meta.	Tremendous hype erupts over the concept of the metaverse: a reenvisioning of the Internet and Web as an immersive 3-D experience.
2022	SpaceX's Starlink takes the lead in the race to deploy low earth orbit (LEO) Internet access satellite systems.	LEO satellite systems can provide broadband Internet access to underserved areas and help bridge the digital divide.

Trends in
E-Commerce Infrastructure 2022 - 2023

BUSINESS
• The Covid-19 pandemic continues to affect businesses of all kinds worldwide, with both short-term and long-term impacts. • Mobile devices provide a rapidly expanding social marketing and advertising platform and foundation for location-based web services and business models. • The growth in cloud computing and bandwidth capacity enables new business models for distributing content and other types of services online. • Big data (huge sets of structured, semistructured, and unstructured data) produced by the Internet creates new business opportunities for firms with the analytic capability to understand it.

Trends in E-Commerce Infrastructure 2022 - 2023

TECHNOLOGY

- The number of Internet users worldwide and amount of time spent online continues to climb, potentially straining Internet and e-commerce infrastructure.
- Mobile devices become the dominant mode of access to the Internet; mobile apps threaten the dominance of the Web.
- Cloud computing reshapes computing and storage and becomes an important force in the delivery of software applications and online content.
- The Internet runs out of IPv4 addresses; the transition to IPv6 continues.
- The decreased cost of storage and advances in database software enables the collection and analysis of big data.
- 5G cellular technology begins to be widely deployed, providing faster Internet access for mobile devices.
- The Internet of Things (IoT), with billions of sensor-equipped devices connecting to the Internet, is powering the development of smart connected "things" such as televisions, houses, cars, and wearable technology.
- Artificial intelligence technologies play an increasing role in e-commerce, with applications ranging from supply chain logistics to intelligent personal assistants to autonomous vehicles.
- The metaverse, a reenvisioning of the Internet and Web as an immersive experience based on virtual and augmented reality technologies, gains traction with Facebook's rebranding as Meta.
- The concept of Web3, a new kind of Internet service premised on blockchain technology and even more decentralized than the current Internet, starts to gain attention.

Trends in
E-Commerce Infrastructure 2022 - 2023

SOCIETY
• The Covid-19 pandemic creates social and ethical challenges, including new security and privacy issues, as well as heightening concerns about the digital divide. • Conflicts among nations impact governance of the Internet. • Government control over, and surveillance of, the Internet expands in many nations. • Countries such as Russia, China, and Iran increasingly disconnect from the global Internet, creating concerns about the Internet ecosystem fracturing into "splinternets": a collection of isolated networks completely controlled by sovereign nations. • The infrastructure for tracking online and mobile consumer behavior conflicts with intensifying individual claims to privacy and control over personal information.

The Internet and The Web : Features and Services

The Internet and the Web have spawned a number of powerful software applications upon which the foundations of e-commerce are built.

You can think of all these as web services, and it is interesting as you read along to compare these services to other traditional media such as television or print media. If you do, you will quickly realize the richness of the Internet environment.

The Internet and The Web : Features and Services

What are the features and services that support E-Commerce?

The Internet and The Web :
Features and Services

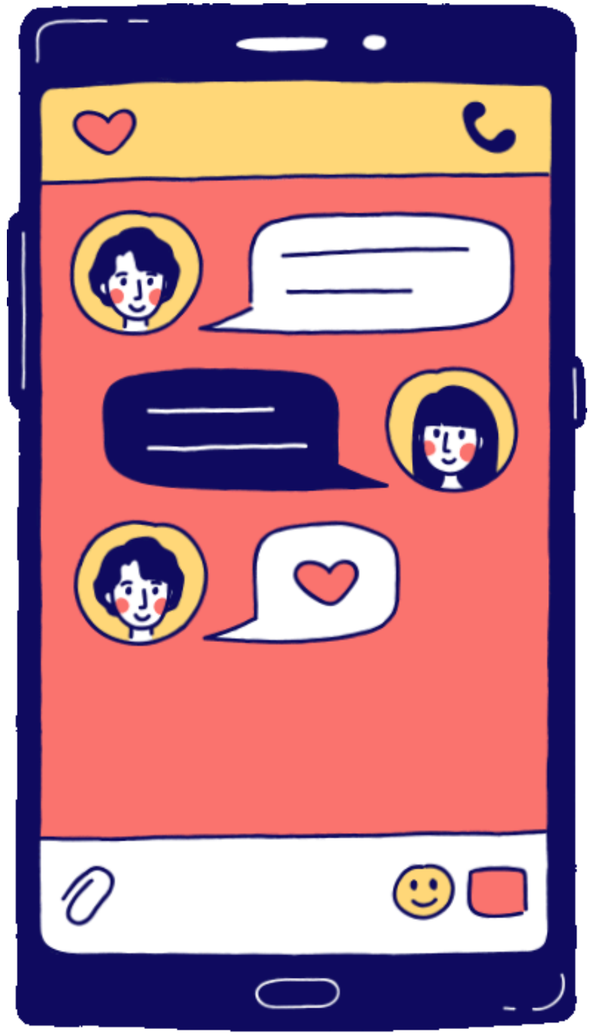
E-Mail

- The most-used application of the Internet. Uses a series of protocols to enable messages containing text, images, sound, and video clips to be transferred from one Internet user to another

The Internet and The Web :
Features and Services

Instant Messaging (IM)

- IM displays text entered almost instantaneously. Recipients can then respond immediately to the sender the same way, making the communication more like a live conversation than is possible through e-mail .



The Internet and The Web :
Features and Services

Online Message Boards /Social Networks/Blogs

- An online message board (also referred to as a forum, bulletin board, discussion board, discussion group, or simply a board or forum) is a web application that enables Internet users to communicate with each other, although not in real time.
- Online social networks are services that support communication within networks of friends, acquaintances, people with similar interests, colleagues, and even entire professions.
- A blog (originally called a weblog) is a personal web page that typically contains a series of chronological entries (newest to oldest) by its author, and links to related web pages.

The Internet and The Web :
Features and Services

Internet Protocol Telephony

- IP Telephony is a general term for the technologies that use Voice over Internet Protocol (VoIP) and the Internet's packet-switched network to transmit voice, fax, and other forms of audio communication over the Internet.

The Internet and The Web :
Features and Services

Videoconferencing, Video Chatting, and Telepresence

- Internet videoconferencing and video chatting is accessible to anyone with a broadband Internet connection and a web camera (webcam).
- Popular videoconferencing tools for businesses include Zoom, Webex (owned by Cisco), Microsoft Teams, Google Hangouts, GoToMeeting, and FUSE. Slack, a cloud-based team collaboration tool that includes videoconferencing and chatting, is another popular option.

The Internet and The Web :
Features and Services

Search Engines

- Search engines identify web pages that appear to match keywords, also called queries, entered by a user, and then provide a list of the best matches (search results).
- Over 85% of U.S. Internet users regularly use search engines from either desktop or mobile devices, and they generate around 18 billion queries a month on desktop computers, about 11.9 billion of which are conducted using Google.

(A) Indexing the Web

1. A Googlebot (software code) crawls the Web, going from link to link.

2. Crawled pages are analyzed: links, semantic analysis, and JavaScript/CSS content.

3. New data is added to an index of keywords and the pages on which they appear.

4. To deal with the scale of the Web (Google has already indexed an estimated 30 trillion pages), Google breaks it up into thousands of "index shards"—groups of millions of pages.

5. The index shards are stored on Google servers (approximately 1 million), located in data centers around the world.

6. The indexing process runs continuously, processing billions of web pages a day. Pages with frequently updated content and links from other highly ranked sites are crawled more regularly and deeply, and given higher rank themselves.

(B) Processing a Search Query

1. A user enters a search query on a desktop computer or mobile device. Google will make suggestions as the user types.

2. The search request is sent to one of Google's many servers.

3. The server uses an algorithm to access the index database, find matching pages, and compute a score, representing how good a match the page is for the query. The algorithm has 200+ variables, including PageRank, the quality and relevance of the content on the page to the query, the context of the search (such as the user's location and device being used), and the user's previous search history. Google also applies various penalties and filters to prevent attempts to "game" the algorithm.

4. Short text summaries (snippets) are generated for each result.

5. Results are delivered to the user, 10 to a page.

The Internet and The Web :
Features and Services

Downloadable, Streaming Media and Podcast

- **Streaming media** is an alternative to **downloaded media** and enables video, music, and other large-bandwidth files to be sent to a user in a variety of ways that enable the user to play the files as they are being delivered.
- A **podcast** is an audio presentation—such as a radio show, audio from a conference, or simply a personal presentation—stored online as a digital media file.

The Internet and The Web :
Features and Services

Virtual Reality and Augmented Reality

- **Virtual reality** (**VR**) involves fully immersing users within a virtual world, typically through the use of a head-mounted display (HMD) connected to headphones and other devices that enable navigation through the experience and allowing users to feel as if they are actually present within the virtual world.
- **Augmented reality** (**AR**) involves overlaying virtual objects over the real world, via smartphones, tablets, or HMDs.
- **Mixed reality** (**MR**) is an enhanced version of augmented reality in which virtual images can interact with the real environments onto which they are overlaid. Microsoft's HoloLens, a head-mounted holographic computer, and the Magic Leap One MR headset are two examples of devices created to enable mixed reality.

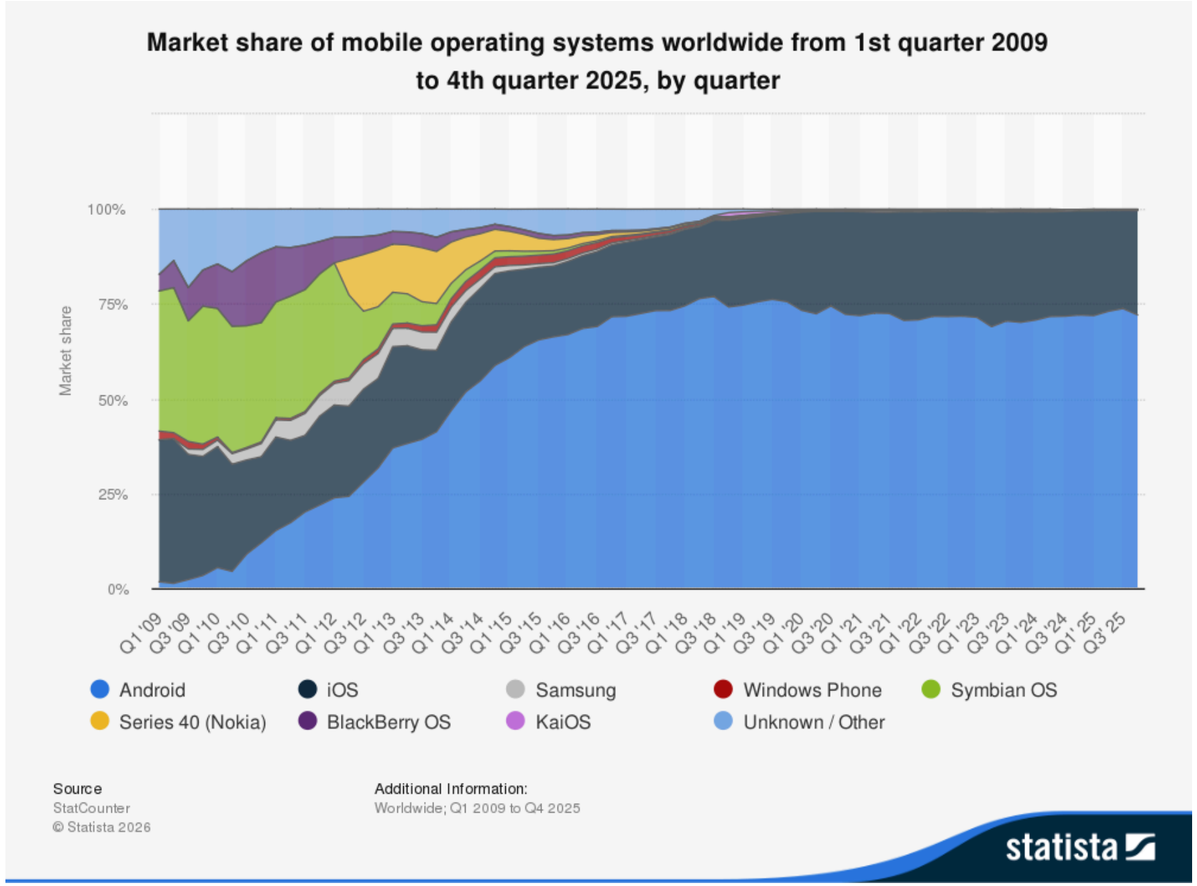
Mobile Apps : The Next Big Thing is Here

When Steve Jobs introduced the iPhone in January 2007, no one, including himself, envisioned that the device would launch a software revolution or become a major e-commerce platform, let alone a game platform, advertising platform, and general media platform for television shows, movies, videos, and e-books.

The implications of the app ecosystem for e-commerce are significant. The smartphone in your pocket or the tablet computer on your lap has become not only a general-purpose computer, but also an always-present shopping tool, as well as an entirely new marketing and advertising platform for vendors.

Mobile Apps : The Next Big Thing is Here

Market share of mobile operating systems worldwide from 1st quarter 2009 to 4th quarter 2025, by quarter

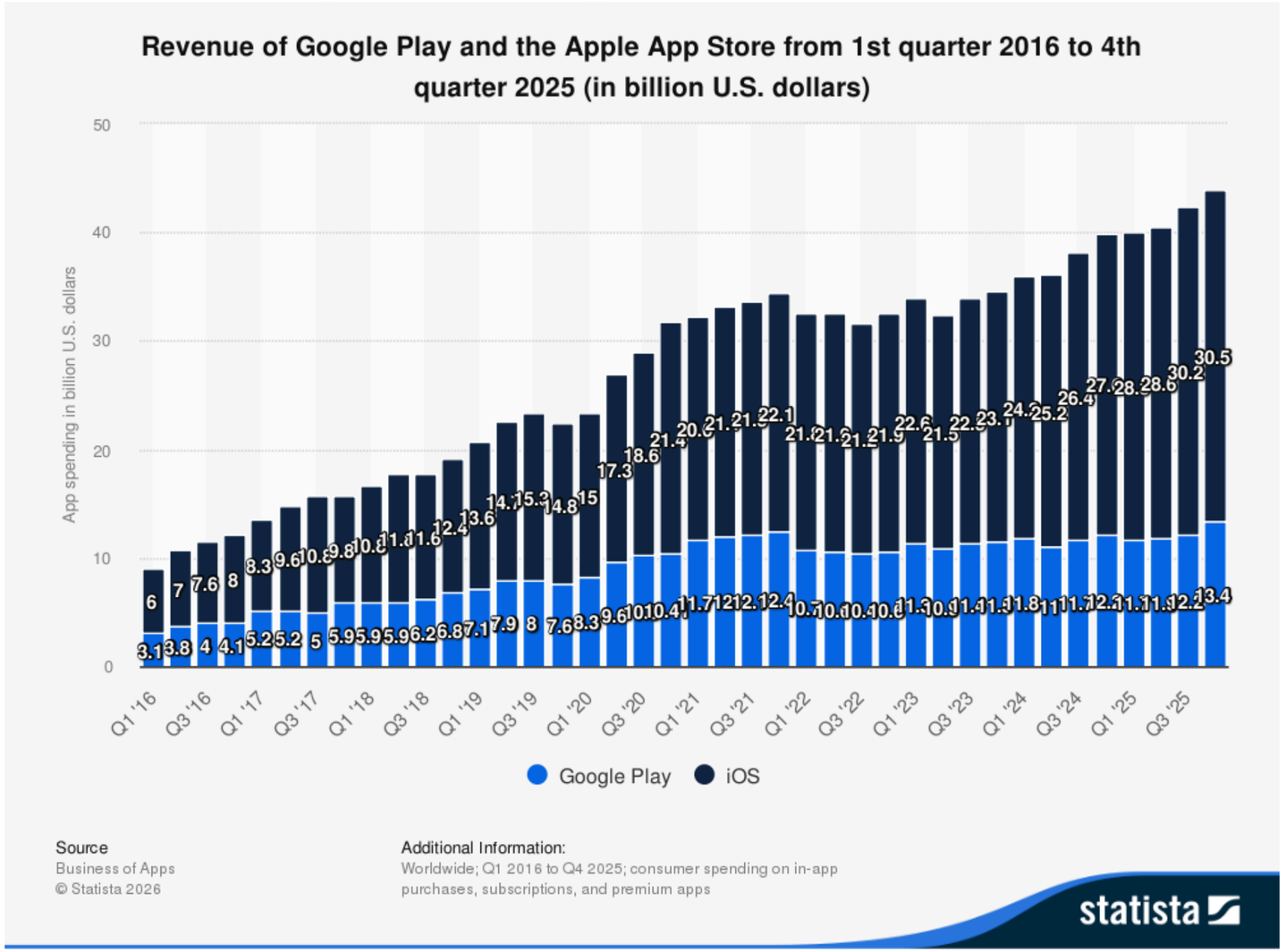


	Q4 '24	Q4 '25
Android	72.04%	72.07%
iOS	27.49%	27.52%
Samsung	0.31%	0.23%
Windows Phone	0.01%	0.01%
Symbian OS	0%	
Series 40 (Nokia)	0%	
BlackBerry OS	0%	
KaiOS	0.03%	0.03%
Unknown / Other	0.01%	0.12%

Source : https://www.statista.com/statistics/272698/global-market-share-held-by-mobile-operating-systems-since-2009/

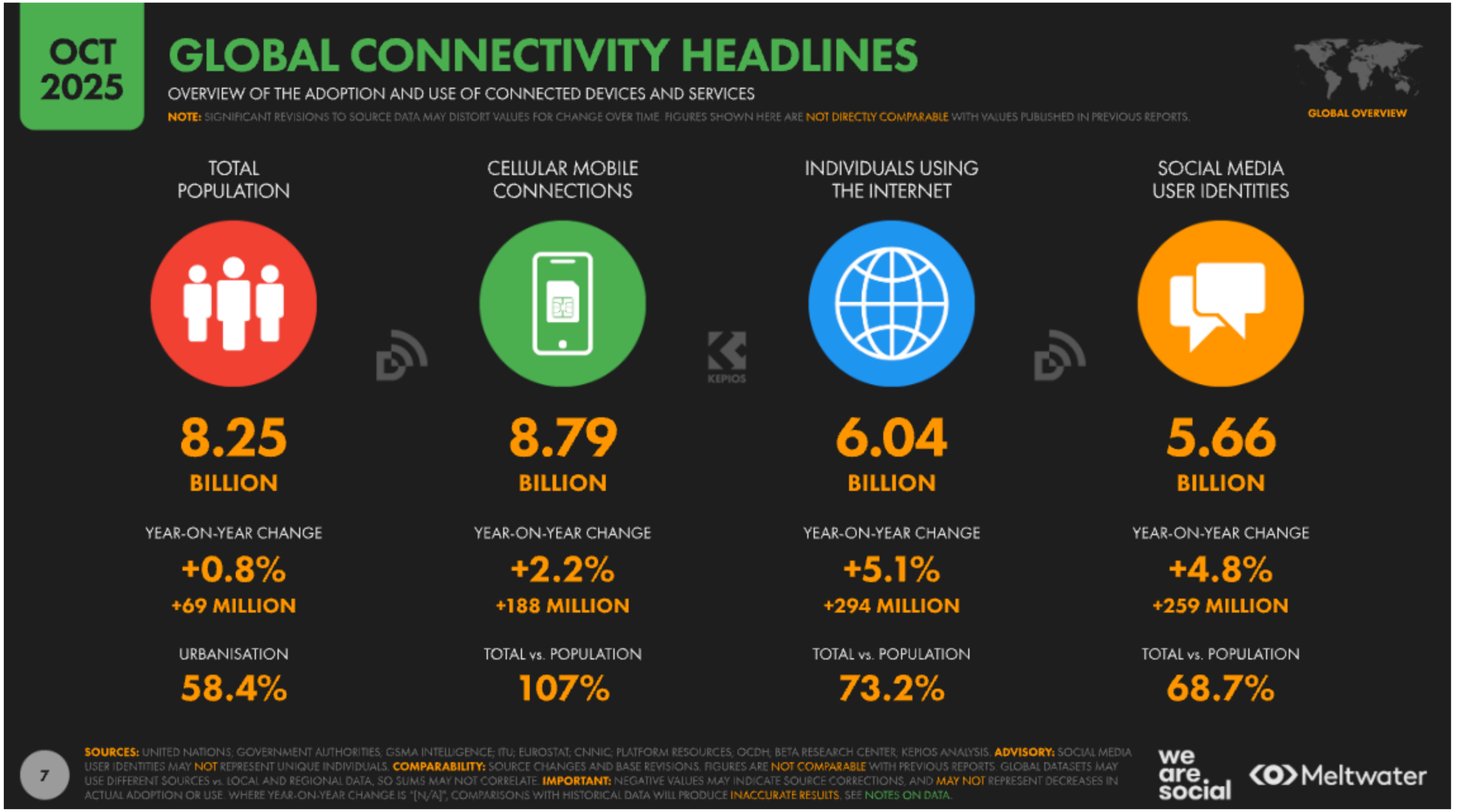
Mobile Apps : The Next Big Thing is Here

Revenue of Google Play and the Apple App Store from 1st quarter 2016 to 4th quarter 2025 (in billion U.S. dollars)



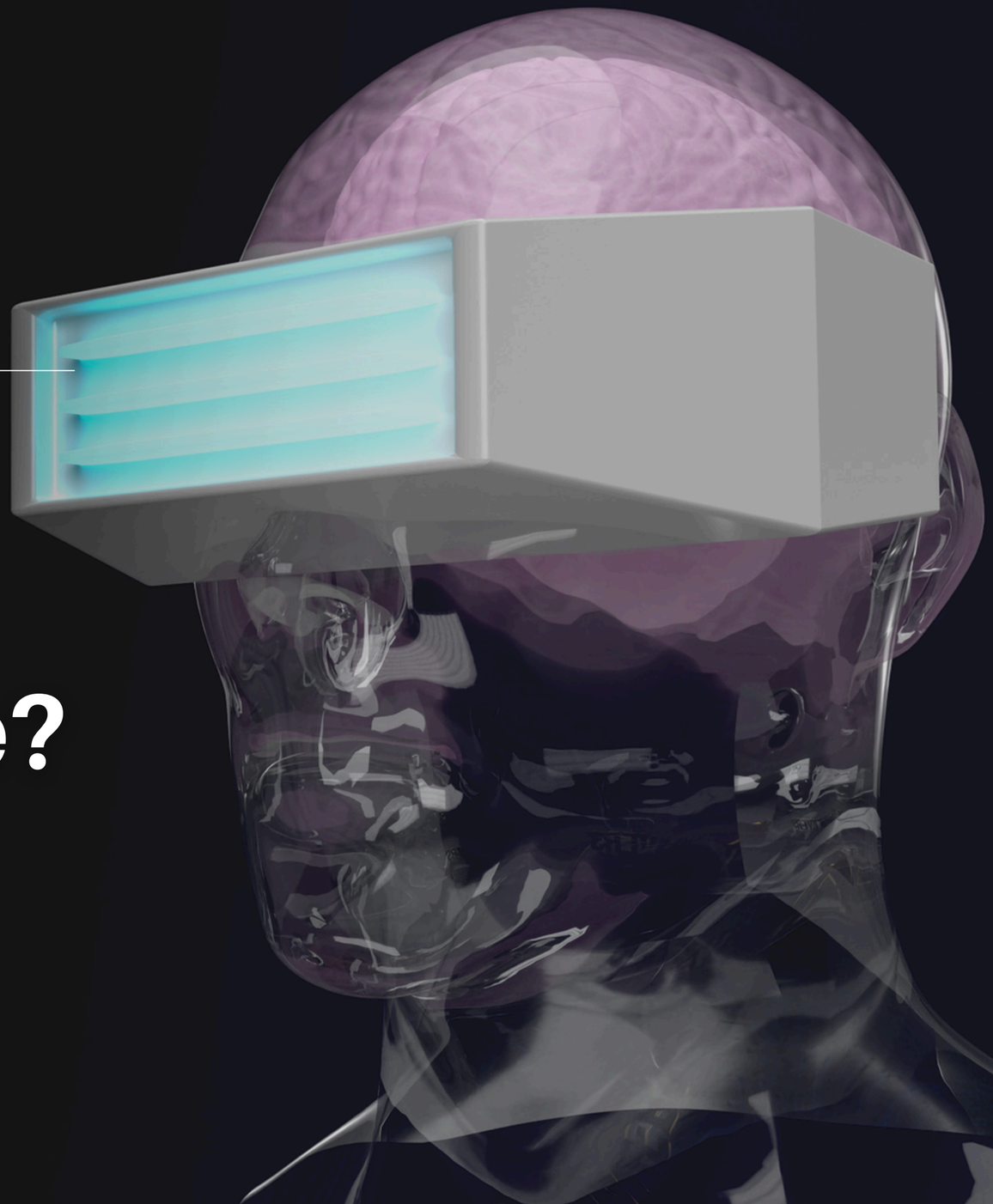
Source : https://www.statista.com/statistics/183469/app-stores-global-revenues/

Mobile Apps : The Next Big Thing is Here



Source : https://datareportal.com/reports/digital-2026-thailand

Could a New Innovation Be a New Infrastructure?



Metaverse/NFT/Cryptocurrency/Blockchain

Walmart : Shopping in the Metaverse

Walmart is quietly preparing to enter the metaverse

News :
https://www.cnbc.com/2022/01/16/walmart-is-quietly-preparing-to-enter-the-metaverse.html

Clip:
https://www.youtube.com/watch?v=u6UYGwPaLf8

In Class Activity 1 :
Building Our Own E-Commerce System VS Using E-Commerce Solution

ให้นักศึกษาอภิปรายร่วมกันในเรื่องของการสร้าง E-Commerce ขึ้นมาใช้เอง หรือการใช้ E-Commerce แบบสำเร็จรูป หรือการขายสินค้าผ่าน Marketplace ผ่านบอร์ดเกม E-Commerce Infrastructure

Board Game : E-Commerce Infrastructure

เป้าหมาย

ออกแบบ E-Commerce Infrastructure ให้เหมาะกับ Business Scenario ภายใต้ Budget 100 Coins แล้วรับมือ Event ที่เปลี่ยนเงื่อนไขระหว่างเกม

กติกาหลัก

1. รับ Scenario Card 1 ใบ และ Infrastructure Deck 1 ชุด
2. เลือก Platform Card (P01-P04) **1 ใบเท่านั้น**
3. เลือก Infrastructure เพิ่มได้จนมีการ์ดรวม **ไม่เกิน 8 ใบ** และ Cost รวมไม่เกิน **100 Coins**
4. วางการ์ดบน Game Board ให้สัมพันธ์กับ Customer Journey และเขียนเหตุผลสั้น ๆ
5. เมื่อเปิด Event Card ทีมสามารถเปลี่ยนการ์ดได้ไม่เกิน 2 ใบ แต่แบบใหม่ยังต้องไม่เกิน 100 Coins
6. Pitch 45 วินาที: Business -> Platform -> 3 Infrastructure สำคัญ -> Event response

Do you have any questions?

มีคำถามหรือข้อสงสัยเกี่ยวกับเนื้อหาที่เรียนในวันนี้

Next Session Topic

หัวข้อสำหรับสัปดาห์หน้า

- Building an E-Commerce Presence